

1 The model

This section fixes theories, calculi, priors, the instantiation grammar, likelihoods and Hänni’s scores. The tracks’ calculi and likelihoods are set side by side in Tables 1 and 2, so that each later result can name the ones it assumes. The results here are structural. Proofs are in Appendix A.

1.1 Syntax and theories

We use the syntax of the earlier report Claude (Anthropic) (2026) (“AS” below), §2, without change. The languages are $L_A = \{0, S, +, \cdot, <, =\}$ and $L_\in$. Formulas may contain *parameters* $p, p_1, \dots$ (free names) and are read under their universal closure; leading universal quantifiers are not stripped, so $\forall x(x + 0 = x)$ and $p + 0 = p$ are different data with equivalent closures (closure-normal form, AS §2.1). Bound variables are de Bruijn indices in the theory and the code; the text writes named variables. $\mathcal{S}$ is the set of these formulas, called *sentences*.

A *template* (AS Def 2.1) may contain metavariable occurrences $M(t_1, \dots, t_n)$ with metavariable-free arguments. Term metavariables are lower case (z, z' if 0-ary; f, g), formula metavariables capital (P, F). An occurrence is a *pattern occurrence* if its arguments are distinct bound variables; in the class DT° every metavariable has one. $\text{FO} \subseteq \text{PAT} \subseteq \text{DT}^\circ \subseteq \text{SO}^\circ$, and DT_F° (0-ary term metavariables) is the class implemented in code. A *ground* template is a single sentence. A substitution θ maps metavariables to bodies, λ -terms whose only free names are parameters, and $\text{inst}(\tau) = \{\tau\theta\}$ (AS Def 2.2); whether parameters are admissible in bodies is said at each use. *Matching* (AS Thm 2.5): for $\tau \in \text{DT}^\circ$ and $s \in \mathcal{S}$ at most one θ has $\tau\theta = s$, found in time $O(|\tau| + |s|)$. Tracks universal and experiments give each metavariable a *guard*: *closed* (no parameter in the body; the default) or *open*.

Robinson arithmetic Q has the axioms Q1 $\forall x \neg Sx = 0$; Q2 $\forall x \forall y (Sx = Sy \rightarrow x = y)$; Q3 $\forall x (\neg x = 0 \rightarrow \exists y x = Sy)$; Q4 $\forall x (x + 0 = x)$; Q5 $\forall x \forall y (x + Sy = S(x + y))$; Q6 $\forall x (x \cdot 0 = 0)$; Q7 $\forall x \forall y (x \cdot Sy = x \cdot y + x)$, numbered as in all four tracks and AS. Track pa writes Q3 as $x = 0 \vee \exists y x = Sy$ and adds $\text{Dlt} := \forall u \forall v (u < v \leftrightarrow \exists z (u + Sz = v))$. Induction is the template $T_{\text{Ind}} = (P(0) \wedge \forall x (P(x) \rightarrow P(Sx))) \rightarrow \forall x P(x) \in \text{DT}^\circ \setminus \text{PAT}$.

Definition 1.1 (Theories). ^{model Def 1.1} A *theory* is a finite set $T = \{\tau_1, \dots, \tau_k\}$ of DT° templates, taken up to renaming of metavariables; $\bigcup \text{inst}(T) := \bigcup_i \text{inst}(\tau_i)$. $\mathcal{H}$ is the class of all theories. A *weighted theory* is a pair (T, w) with w in the open simplex over T .

1.2 Calculi

Definition 1.2 (Background calculus; theorems). ^{model §1.2} The background calculus is Mendelson’s first-order calculus with equality K (Mendelson, 1997, Ch. 2; chapter and numbering not checked): axiom schemas A1 $B \rightarrow (C \rightarrow B)$; A2 $(B \rightarrow (C \rightarrow D)) \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow D))$; A3 $(\neg C \rightarrow \neg B) \rightarrow ((\neg C \rightarrow B) \rightarrow C)$; A4 $\forall x B \rightarrow B[t/x]$, t free for x ; A5 $\forall x (B \rightarrow C) \rightarrow (B \rightarrow \forall x C)$, x not free in B ; reflexivity and substitutivity $x = y \rightarrow (B(x, x) \rightarrow B(x, y))$ of equality; rules MP and Gen. Mendelson’s free variables are our parameters; Gen infers $\forall x \varphi[x/p]$ from φ . $\text{Th}(T)$ is the set of formulas derivable in K from $\bigcup \text{inst}(T)$, with Gen on every parameter; $\text{Th}_d(T)$ the set of those with a derivation of symbol size at most d , axioms included.

Since parameters are read under closure, $\varphi(p)$ and $\forall x \varphi$ are interderivable (AS Prop 4.10), while a closed instance $\varphi(t)$ is not (AS Ex 4.9). K is complete (known). $\text{Th}(T)$ and $\text{Th}_d(T)$ depend on T only through $\bigcup \text{inst}(T)$. The unit of d is stated at each use: symbol size here and in ??, chain steps in ??, derivation size in ??.

calculus	rules	normalisation	factor $P_{H_\forall}/P_{H_{\text{sch}}}$ per closed instance	used in
K	cite theory or logical axiom, MP, Gen, $\forall E$	μ_T/Z_T ; finite a.s. iff $m \leq 1$	not computed; with parameters admissible no hypothesis emits only instances (??(c3))	§§??–??
$C_{\min}$	cite $(1 - c)$, $\forall E (c)$	μ_T or μ_T/Z_T	c unnormalised; $c/(1 + c)$ normalised (φ quantifier-free)	§??
C_{open}	$C_{\min} + \text{Gen on } p (g)$	same	$c(1 - \rho)/(1 + c)$ normalised, against guarded H_{sch} (ρ : parameter probability of $\mathcal{Q}_{\text{open}}$)	§??
C_{U3}	cite, $\forall E$, quantifier-free rules	trees, μ_T	$\leq c$ on the mass that cites the axiom (??)	§??
$C_\wedge$	cite, $\forall E$, $\wedge I$, $\wedge E$	trees, μ_T	$> c$ (refuted conjecture, ??); ≤ 0.925 on the computed grid	§??
Ch_J	cite, then $\leq J$ steps ($\forall E$, Gen, MP)	proper, no normaliser	$c_{\text{ch}} = 1 - c_{\text{stop}}$	§??
ND	hyp, ax, tc, $\rightarrow I$, $\forall E$, $\forall I$, $\exists I$, $\exists E$, refl, subst	two-part code; sub-probability	1/10 per $\forall E$ step (fixed rule code; learned codes: ??)	§??
L1^{eq}	cite, refl, sym, trans, cong _S , cong ₊	μ_T/Z_T , least fixpoint	— (closed equations only)	§??

Table 1: The calculi of the tracks. “Factor” is the likelihood ratio of $H_\forall = \{\forall x\varphi\}$ to its instance schema H_{sch} on one closed instance $\varphi(t)$. Rates are comparable only within one calculus and one code: Ch_J is proper, yet its factor equals the *unnormalised* $C_{\min}$ factor with $c = c_{\text{ch}}$. The direction (factor at most 1) is proved for $C_{\min}$ and C_{U3} (??), follows from the exact laws for C_{open} (??), and holds in Ch_J (experiments Prop X6) and under pa’s codes (pa §5.5); in $C_\wedge$ it holds on the computed grid and is open in general.

Lemma 1.3 (A4 is not a template). *[(a) proved; (b) proof sketch]^{model Lemma 1.2} (a) $A4$, written as $\forall xP(x) \rightarrow P(f)$ with f a 0-ary term metavariable, is not in SO° , so not in DT° . (b) No finite set of DT° templates has the instances of $A4$ as its instance union.*

Proof idea. (a) AS Def 2.1 requires metavariable-free arguments. (b) A template inside $A4$ covers $\forall xB_m \rightarrow B_m[t/x]$, $B_m := (x + \dots + x = x)$, for one m or one t (Appendix A.3). $\square$

So $\forall$ -elimination belongs to the fixed background (compare AS Rem A.3). The other logical axioms are templates: A1–A3 in FO, A5 in PAT as $\forall x(P \rightarrow F(x)) \rightarrow (P \rightarrow \forall xF(x))$, substitutivity in DT° . Derivation grammars use $\forall E$ as a primitive rule; it is derivable in K, so Th is unchanged.

Definition 1.4 (The calculi of the tracks). ^{universal §1.4; experiments §1.6; pa Def 0.3; model Ex 3.6} Each track’s derivation likelihood is a stochastic grammar over derivations in its own calculus (Table 1; rule lists in Appendix A.3): track universal’s chains $C_{\min}$ (cite with probability $1 - c$, $\forall E$ with probability c) and $C_{\text{open}}(c, g)$ (plus Gen on the parameter with probability g ; $p_{\text{cite}} = 1 - c - g$), and its tree calculi C_{U3} and $C_\wedge$; track experiments’ forward chain Ch_J (stop probability c_{stop} while a rule applies; $c_{\text{ch}} := 1 - c_{\text{stop}} = \frac{1}{2}$); track pa’s natural deduction ND, scored by the two-part code L1^{sch} ; and the equational grammar of L1^{eq} .

1.3 Prior

Definition 1.5 (Template code and prior). ^{model Def 1.3} Templates are coded in preorder, one token per node from a fixed alphabet (the symbols of the language, connectives, quantifiers, and escapes IDX, PAR, MV) at $\lceil \log_2(\text{alphabet size}) \rceil$ bits. The escapes are followed by Elias- γ codes:

$\gamma(k+1)$ for a de Bruijn index k , $\gamma(j+1)$ for p_j , and for a metavariable a new/old bit, then its arity ($\gamma(n+1)$) and sort bit if new or its index if old, then its arguments. Renamings get the same code, argument permutations ($M(x, y)$, $M(y, x)$) different ones. A theory $\{\tau_1, \dots, \tau_k\}$ is coded as $\gamma(k+1)$ and the k template codes in lexicographic order; $\ell(T)$ is the length in bits, and

$$\pi_\lambda(T) := 2^{-\lambda\ell(T)} / Z_\lambda, \quad \lambda \geq 1, \quad Z_\lambda := \sum_{T \in \mathcal{H}} 2^{-\lambda\ell(T)}.$$

Mixture weights get the prior $\text{Dir}(\alpha, \dots, \alpha)$, by default $\alpha = \frac{1}{2}$ (Krichevsky and Trofimov, 1981). Statements that need fixed weights say so.

Lemma 1.6 (Kraft). *[proved]^{model Lemma 1.4} $\sum_{T \in \mathcal{H}} 2^{-\ell(T)} \leq 1$, so π_λ is a probability for $\lambda \geq 1$; and $\sum_T \pi_\lambda(T)^{1/2} < \infty$ for $\lambda \geq 2$.*

Proof idea. Theory codes are prefix-free (Appendix A.1); for $\lambda \geq 2$, $2^{-\lambda\ell/2} \leq 2^{-\ell}$. $\square$

Remark 1.7 (Other priors; time factors). *[proved]^{experiments §1.3; universal §1.6; pa Def 0.2; model §1.3} (i) Track experiments' prior $\propto 2^{-\lambda \text{bits}(T)} (1 + |T|)^{-\tau}$ uses a stochastic template code that satisfies Kraft, so it is proper for $\lambda \geq 1$ ($\lambda = \frac{1}{2}$ only on finite pools); $\mathbb{Q} \cup \{T_{\text{Ind}}\}$ costs 226.2 bits. Its factor $(1 + |T|)^{-\tau}$ (time exponent τ , default 0) prices membership time. (ii) Track universal's $2^{-\lambda(\sum_A |A| + \#\text{axioms})}$ and track pa's $2^{-\beta \sum_\tau |\tau|}$ ($\beta = 5$ or $\log_2 23$ bits per symbol) are improper over all theories and are used only on finite or summable families. Prior shares depend on the code (??). (iii) A Levin-style time factor for a decider of the axiom set is polynomial in $\ell(T)$ inside DT° , since membership is matching; it changes $\ln \pi$ by $O(\ln \ell(T))$ and barely bites. Time matters in ??.*

1.4 Instantiation grammar

Definition 1.8 (Instantiation grammar $\mathcal{Q}$). ^{model Def 1.5} $\mathcal{Q}$ is a probabilistic context-free grammar on metavariable bodies with one production law per *type* (s, k) : the sort $s \in \{\iota, o\}$ of the node and the number k of variables available there (holes plus bound indices in scope; the two are treated alike). A body of type $\iota^n \rightarrow s$ is generated from the root type (s, n) . A node of type (s, k) chooses, with probabilities depending only on (s, k) , a function or relation symbol or connective (children of type $(\cdot, k)$), a quantifier (child of type $(o, k+1)$), one of the k variables, a constant, or a parameter (from a geometric law). $\mathcal{Q}(\theta) := \prod_M \mathcal{Q}_{\text{type}(M)}(\theta(M))$. We require *full support* (all productions have positive probability) and *uniform subcriticality*: at most $b < \infty$ children per production, and weights $h(\text{type}) \in [h_{\min}, h_{\max}]$, $h_{\min} > 0$, and $\rho < 1$ with $\mathbb{E}_i[\sum_{\text{children } c} h(\text{type}(c))] \leq \rho h(i)$ for every type i .

Lemma 1.9 ($\mathcal{Q}_\tau$ is a law on $\text{inst}(\tau)$). *[proved]^{model Lemma 1.6} Let $\mathcal{Q}$ be uniformly subcritical, $\tau \in \text{DT}^\circ$, and N the size of a body drawn from type i . (a) $\mathbb{E}_i N \leq h(i)/(h_{\min}(1 - \rho))$; bodies are finite a.s. (b) With $\Lambda := 2/((1 - \rho)h_{\min})$ and $0 < \zeta \leq (\Lambda b h_{\max})^{-2}$: $\mathbb{E}_i[e^{\zeta N}] \leq e^{\zeta \Lambda h(i)}$, so $P_i(N \geq \ell) \leq e^{\zeta(\Lambda h_{\max} - \ell)}$. (c) $\mathcal{Q}_\tau(s) := \mathcal{Q}(\theta_s)$ if $\tau\theta_s = s$, else 0, is a probability on $\mathcal{S}$; with full support its support is $\text{inst}(\tau)$. (d) (a) and (b) hold for every multi-type branching process with independent children given the type, at most b children and the uniform condition.*

Proof idea. The h -weight of generation j has mean at most $\rho^j h(i)$; (b) is an induction on depth; (c) uses unique matching (Appendix A.2). $\square$

Remark 1.10 (The earlier condition). *[refuted (old condition); exclusion proved; computed]^{model §1.4} The pre-referee condition “every type has mean offspring below 1” is refuted (referee M5) as a usable*

hypothesis: no formula type of L_A satisfies it, and with infinitely many types it does not give finiteness (a chain infinite with probability $\prod_k (1 - 1/(k+2)^2) = \frac{1}{2}$). The uniform condition excludes that chain and holds for a grammar for L_A with $\rho = 0.75$ (Appendix A.2).

The term laws $\mathcal{Q}_{\text{num}}$, $\mathcal{Q}_{\text{GW}}$, $\mathcal{Q}_{\text{open}}$ of track universal (??) and track experiments' grammar (Example A.1) are of this kind. Track pa instead *learns* a positional grammar (a sequential KT model of the body). It is not a fixed PCFG, so results that assume a fixed $\mathcal{Q}$, such as the exact split ties of ??, do not apply to it verbatim (??).

1.5 Likelihoods

Data are a sequence $D = (X_1, \dots, X_n)$ of sentences.

Definition 1.11 (L0: axiom citation). ^{model §1.5} With fixed weights, $P_{T,w}(s) := \sum_{\tau \in T} w_\tau \mathcal{Q}_\tau(s)$ and $P_{T,w}(D) := \prod_i P_{T,w}(X_i)$. With Dirichlet weights, $P_T^{\text{Dir}}(D) := \int \prod_i P_{T,w}(X_i) \text{Dir}(dw; \alpha)$.

Lemma 1.12 (Dirichlet labelling sum). *[proved; computed]^{model §1.5 (c8); experiments Prop X2} With $A := \sum_\tau \alpha_\tau$, summing over labellings $\ell : \{1..n\} \rightarrow T$ with $X_i \in \text{inst}(\ell_i)$, and $n_\tau(\ell)$ the number of data labelled τ ,*

$$P_T^{\text{Dir}}(D) = \sum_\ell \frac{\Gamma(A)}{\Gamma(A+n)} \prod_{\tau \in T} \frac{\Gamma(\alpha_\tau + n_\tau(\ell))}{\Gamma(\alpha_\tau)} \prod_{i=1}^n \mathcal{Q}_{\ell_i}(X_i).$$

P_T^{Dir} is exchangeable and consistent in n , and its normaliser does not depend on D . For disjoint instance sets the sum has one term, and $P(X_{n+1} = s \mid D) = \sum_\tau \frac{\alpha_\tau + n_\tau}{A+n} \mathcal{Q}_\tau(s)$, the KT rule per template. (Numerical checks: Appendix A.4.)

Definition 1.13 (L1: derivation grammar). ^{model §1.5} Fix rule probabilities $\alpha_{\text{ax}}, \alpha_{\text{lg}}, \alpha_{\text{MP}}, \alpha_{\text{Gen}}, \alpha_{\text{VE}}$ summing to 1, with $\alpha_{\text{ax}} + \alpha_{\text{lg}} > 0$; put $\alpha_r := \alpha_{\text{MP}} + \alpha_{\text{Gen}} + \alpha_{\text{VE}}$ and $m := 2\alpha_{\text{MP}} + \alpha_{\text{Gen}} + \alpha_{\text{VE}}$. A tree is grown from the root; each node independently cites a theory axiom (α_{ax} : τ with probability w_τ , then $\theta \sim \mathcal{Q}$; conclusion $\tau\theta$), cites a logical axiom (α_{lg} : a schema of K uniformly, instantiated by $\mathcal{Q}$), or applies MP (two children), Gen (one child; a parameter from a geometric law is abstracted) or $\forall E$ (one child; a term $t \sim \mathcal{Q}$). A tree is *valid* if at each MP node the right child concludes (left conclusion $\rightarrow B$), the node concluding B , and each $\forall E$ node's child has root $\forall$. Then $\mu_T(s) := \Pr[\text{valid, conclusion } s]$, $Z_T := \sum_s \mu_T(s)$ and $P_T(s) := \mu_T(s)/Z_T$.

The code length $-\log_2 \Pr(\text{tree})$ is a derivation-length prior that counts *grammar choices*, not the symbols of the formulas; the two can differ exponentially (??).

Lemma 1.14 (L1 is proper; support). *[(a) proved, except the extinction criterion (known); (b), (c) proved; computed]^{model Lemma 1.7} (a) Let $\mathcal{Q}$ be uniformly subcritical. The tree is finite a.s. iff $m \leq 1$ (Athreya and Ney, 1972, Ch. I; numbering not checked); for $m < 1$ it has $1/(1-m)$ derivation nodes in expectation; for $m > 1$ it is infinite with probability $1 - q > 0$, q the least root of $q = (\alpha_{\text{ax}} + \alpha_{\text{lg}}) + (\alpha_{\text{Gen}} + \alpha_{\text{VE}})q + \alpha_{\text{MP}}q^2$. (b) $Z_T \geq 1 - \alpha_r > 0$, so P_T is a probability whose normaliser does not depend on the data. (c) If all rule probabilities and weights are positive, $\mathcal{Q}$ has full support and parameters are admissible (in logical-axiom instances, $\forall E$ terms and Gen), then $\text{supp } P_T = \text{Th}(T)$. Without parameters Gen is vacuous, and for example $\forall x(R(x) \rightarrow R(x))$ has no tree. Mean sizes and the extinction probability are checked numerically (Appendix A.4).*

Definition 1.15 (Variants). ^{model §1.5; universal §1.5; pa Def 0.3; experiments §1.7; model Ex 3.6}

- L2: L1 with citation forced at depth d ; support $\text{Th}_{\leq d}(T)$, the conclusions of valid trees of depth $\leq d$.

- $L1^\sigma$: $\mu_T^\sigma(s) := \sum_{\pi \text{ valid, concluding } s} \Pr(\pi) e^{-\kappa|\pi|}$, $\kappa > 0$, $|\pi|$ the total symbol size of the tree's conclusions; $P_T^\sigma := \mu_T^\sigma/Z_T^\sigma$.
- $L1^{\max}$ (two-part, Viterbi): $V_T(s) :=$ the largest probability of a single valid tree concluding s (under $L0$: of a single citation).
- $L1^{\text{sch}}$ (pa; ??): a computable two-part code over ND derivations with β bits per *written symbol* (Appendix A.3); reported values are upper bounds. It is a two-part form of $L1^\sigma$, not of plain $L1$. $L1^{\text{naive}}$ writes each formula out at the instance.
- $L0^{\text{cl}}$ (universal): as $L0$, but a universal sentence $\forall \bar{x} \chi$ is cited through its instances $\chi(\bar{t})$.
- $L1^{\text{sel}}$ (stream filter; universal): $P_T(s | S) = P_T(s)/P_T(S)$ for $s \in S$, a set known to contain every datum.
- $L1_{\text{cit}}^{\text{sel}}$ (per-citation rejection; experiments): $\sum_i w_i a_i(s)/c_i$, with $a_i(s)$ the probability that a chain from component i ends in s and c_i that it ends in S .
- Noise mixtures (universal): $P_T^\eta := (1 - \eta)P_T + \eta N$, N a fixed full-support law, $\eta \in (0, 1)$.
- L_ε (pa): $(1 - \varepsilon)P_T(s) + \varepsilon \mu_0(s)[T \not\vdash_k \neg s]$, μ_0 fixed, $\not\vdash_k$ a bounded refutation search; a sub-probability whose mass depends on T , not on the data.
- $L1^{\text{eq}}$ (model Ex 3.6): μ_T/Z_T for the equational grammar of Table 1, terms of size $\leq N$.

Lemma 1.16 ($L1^\sigma$ is proper). *[proved]^{model §1.5} $0 < Z_T^\sigma \leq Z_T \leq 1$; so $L1^\sigma$ is a probability with a data-independent normaliser and the support of $L1$.*

Remark 1.17 (Two selection-aware likelihoods). *[proved]^{experiments §1.7 (referee m8)} $L1^{\text{sel}}$ and $L1_{\text{cit}}^{\text{sel}}$ coincide for one-component theories and differ for mixtures ($\sum_i w_i a_i / \sum_i w_i c_i$ against $\sum_i w_i a_i / c_i$). ?? uses $L1^{\text{sel}}$; experiments E1 and E6 use $L1_{\text{cit}}^{\text{sel}}$, for which multi-component pool members are a modelling choice (??).*

Table 2 in the appendix lists all these likelihoods with their normalisers, whether they have the size principle (Section 1.7), and each track's name for them.

1.6 Hänni's scores

Hänni's working notes (Hänni, 2026)¹ keep the axiom systems that prove the givens “and not their negations”, or those “that do not contradict the given statements”, and suggest that the weight “depend [...] on how many of the given statements can be proven. (one could also penalize longer proofs i guess?)”.

Definition 1.18 (Hänni's scores). ^{model §1.5; universal §1.5; pa Def 0.3} For labelled data $D = D_+ \cup D_-$ (true, false): $S_{\text{prove}}(T; D) := 1[T \text{ consistent}] \prod_{s \in D_+} 1[T \vdash s] \prod_{s \in D_-} 1[T \vdash \neg s]$; $S_{\text{nc}}(T; D) := 1[T \cup D_+ \cup \neg D_- \text{ consistent}]$; $S_g(T; D) := S_{\text{nc}}(T; D) \prod_{s \in D_+} g(\ell_T(s))$, with $\ell_T(s)$ the size of a shortest derivation (∞ if none), g non-increasing and $g_\infty := g(\infty) \in [0, 1]$. The posterior is $\pi(T)S(T; D)$. These are scores, not likelihoods: $\sum_D S(T; D) \neq 1$ and depends on T . Track universal's $S_{\text{nc}, \beta}$ equals $\beta^n S_g$ with $g \equiv 1$ on provable data and $g_\infty = 1/\beta$. Track pa's L_ε is the noisy form: $\varepsilon = 1$ gives S_{nc} up to the factor $\mu_0(D)$, common to all theories, and $\varepsilon \rightarrow 0$ moves towards S_{prove} .

¹File `research/prior/hanni-solomonoff-axiom-induction.md` of the accompanying repository.

1.7 The size principle

Lemma 1.19 (Size principle, exact form). *[proved; computed]^{model Lemma 1.8 (c1); universal Thm U4(1)} Let P^*, P be probabilities on a countable $\mathcal{S}$, $\varepsilon := P(\mathcal{S} \setminus \text{supp } P^*) < 1$ and $\tilde{P} := P(\cdot \mid \text{supp } P^*)$. (a) $\text{KL}(P^* \| P) = \text{KL}(P^* \| \tilde{P}) + \ln \frac{1}{1-\varepsilon} \geq \ln \frac{1}{1-\varepsilon}$. (b) On data in $\text{supp } P^*$, $P(D) = (1-\varepsilon)^n \tilde{P}(D)$, and $\mathbb{E}_{P^*}[\tilde{P}(D)/P^*(D)] = 1$. Computed: for $\{z+0=z\}$ against $\{z_1+0=z_2\}$, (a) holds to 10^{-9} , and the simulated per-datum log-ratio 3.94 nats matches $\text{KL} = 3.894$ nats.*

A hypothesis that puts mass ε outside the data's support loses at least $\ln \frac{1}{1-\varepsilon}$ nats per datum: the size principle (Tenenbaum and Griffiths, 2001). ?? states it for sub-probabilities.

Proposition 1.20 (Which likelihoods have the size principle). *[proved]^{model Prop 1.9} (a) L0 , L1 , L2 and L1^σ have it: each P_T is a probability on $\mathcal{S}$ with a data-independent normaliser, so Lemma 1.19 applies to every T that puts mass outside the data law's support (under L0 with full-support $\mathcal{Q}$: every T with $\bigcup \text{inst}(T) \not\subseteq \text{supp } P^*$). (b) Hänni's scores do not. Let $\text{Th}(T) \subseteq \text{Th}(T')$. Then $S_{\text{nc}}(T') \leq S_{\text{nc}}(T)$. If T' is consistent with the labelled data, $S_{\text{prove}}(T') \geq S_{\text{prove}}(T)$; if moreover $\bigcup \text{inst}(T) \subseteq \bigcup \text{inst}(T')$, $S_g(T') \geq S_g(T)$. So if $\psi \notin \text{Th}(T^*)$ and $T^* \cup \{\psi\} \cup D_+ \cup \neg D_-$ is consistent, the posterior odds of $T^* \cup \{\psi\}$ against T^* never fall below the prior odds under S_{prove} or S_g , and equal them under S_{nc} . (c) Generative likelihoods penalise that strengthening (ψ ground): exactly by $(1-w_\psi)^n$ under L0 with fixed weights, on data not containing ψ ; at rate at least $\ln \frac{1}{1-\alpha_{\text{ax}} w_\psi}$ nats per datum under L1 ; like $n^{-\alpha}$ with Dirichlet weights (??).*

The consistency hypothesis in (b) is needed ($\psi = \forall x R(x)$, $T^* = \emptyset$, $D_+ = \{\neg R(0)\}$; referee m9).

Remark 1.21 (Hänni's graded score, normalised). *[proved; refuted (old sentence); computed]^{model Rem 1.10} Let $\ell_T(s)$ be the least length in bits of an explicit derivation of s from T in a prefix-free code that writes every formula out, $g(\ell) := 2^{-\kappa \ell}$, $\kappa \geq 1$, $g_\infty = 0$. Then $Z_T := \sum_s 2^{-\kappa \ell_T(s)} \leq 1$ and $P_T^g := 2^{-\kappa \ell_T} / Z_T$ is a probability; with $g_\infty > 0$ the normaliser is infinite. Like L1^σ , P_T^g charges the *written* length of the derivation, while L1 charges only grammar choices (??). If $\bigcup \text{inst}(T) \subseteq \bigcup \text{inst}(T')$, then $Z_{T'} \geq Z_T$, so $P_{T'}^g(s) \leq P_T^g(s)$ whenever $\ell_{T'}(s) = \ell_T(s)$. Hänni's remark that one can “think of this as there being some model for how the statements get generated” holds once the score is divided by Z_T and $g_\infty = 0$. *Refuted* (referee m3): the pre-referee sentence “a stronger theory has more short theorems, hence a larger Z_T , hence a smaller $P_T(s)$ on each datum”. For $T = \{a, a \rightarrow b\}$ and $T' = T \cup \{b\}$ (same theorems, nested instances) and $\kappa = 1$, $P_T(b) = 0.032$ and $P_{T'}(b) = 0.346$ (10 sentences gain, 36 lose); under L2 , 0.016 and 0.24. A datum whose shortest derivation gets shorter can gain.*

1.8 Computing the likelihood

Proposition 1.22 (What can be computed). *[(a) proved; (b), (c) proved, given Church's theorem (known)]^{model Prop 2.7} Assume L1 with $m < 1$, $\mathcal{Q}$ uniformly subcritical, computable grammar probabilities and weights, and the hypotheses of Lemma 1.14(c). (a) $\mu_T(s)$, Z_T and $P_T(s)$ are computable reals, uniformly in (T, s) . (b) $\{(T, s) : P_T(s) > 0\} = \{(T, s) : s \in \text{Th}(T)\}$ is recursively enumerable and, for a language with a binary relation symbol, undecidable already for $T = \emptyset$ (Church, 1936; Turing, 1936). (c) No algorithm decides whether a theory has likelihood 0 on a datum. For a finite class, the posterior $\pi_n(T)$ is a computable real whenever it is defined, but which theories it has excluded is undecidable.*

Proof idea. (a) The whole tree, bodies included, satisfies Lemma 1.9(d); finitely many trees have at most N grammar nodes, the rest have probability $O(1/N)$, and $Z_T \geq 1 - \alpha_r$. (b) Lemma 1.14(c) and Church’s theorem (Appendix A.6). $\square$

Cost. Precision ϵ needs all trees with up to $N \approx C_\nu/\epsilon$ grammar nodes (C_ν bounds their mean number), exponentially many in N ; a likelihood ratio needs ϵ below likelihoods that are exponentially small in the datum’s least tree size ($??$). No better general algorithm and no lower bound are known. The tracks restrict the calculus (Ch_J , $J \leq 2$) or score explicit derivations (L1^{sch}).

Proposition 1.23 (The maximum is not the sum). *[proved]^{model Prop 2.8} Let $\tau := \forall x \forall y F(x, y)$ ($F : \iota^2 \rightarrow o$), $\tau' := \forall x \forall y F(y, x)$, $T_1 := \{\tau\}$, $T_2 := \{\tau, \tau'\}$ with weights $\frac{1}{2}, \frac{1}{2}$, and $\mathcal{Q}$ symmetric in the holes. Under L0 , $P_{T_2} = P_{T_1}$, so the posterior odds equal the prior odds at every n . Under L1^{max} , $V_{T_2}(s) = \frac{1}{2}V_{T_1}(s)$ on every instance, so the two-part odds are $(\pi(T_2)/\pi(T_1))2^{-n} \rightarrow 0$: the maximum breaks a tie inside the generator class.*

Proof idea. $\mathcal{Q}_\tau = \mathcal{Q}_{\tau'}$, and each instance has one citation per template (Appendix A.6). $\square$

Here the theorems agree; whether the maximum can change a *deductive* limit is open. Track universal’s L1^{max} and track pa’s codes are of this two-part kind.

A The model: proofs and details

Scripts cited here are in `research/tracks/model/checks/` (the track’s) and `.../referee_code/` (its referee’s); each is seeded and writes its output next to itself.

A.1 Prior

Proof of Lemma 1.6. The template code is prefix-free: each token fixes the number of children that follow, the escapes are followed by self-delimiting Elias- γ codes and fixed bits, and an old metavariable’s arity is already known. $\gamma(k+1)$ fixes how many template codes follow, so theory codes are prefix-free. The code determines a template up to renaming of metavariables, which is how Definition 1.1 identifies templates, and sorting makes the code of a set canonical; so distinct theories have distinct codes, and Kraft’s inequality gives $\sum_T 2^{-\ell(T)} \leq 1$. Hence $0 < Z_\lambda \leq 1$ for $\lambda \geq 1$. For $\lambda \geq 2$, $\sum_T \pi_\lambda(T)^{1/2} = Z_\lambda^{-1/2} \sum_T 2^{-\lambda \ell(T)/2} \leq Z_\lambda^{-1/2} \sum_T 2^{-\ell(T)} < \infty$. $\square$

Implicit in the notes. (1) Theories must be taken up to renaming of metavariables: renamings share a code, so otherwise infinitely many theories would share one code and the sum would diverge. (2) The notes code the number of templates as $\gamma(k)$; Elias- γ codes positive integers, and the empty theory (used in Proposition 1.22(b)) needs a code, so we write $\gamma(k+1)$. Neither changes a statement.

Proof of Remark 1.7. (i) The experiments’ template code is $-\log_2$ of a subcritical stochastic grammar over templates whose arities fix the tree shape, so $\sum_\tau 2^{-\text{bits}(\tau)} \leq 1$. A set of m distinct components costs $|\gamma(m)| + \sum_i \text{bits}(\tau_i) - \log_2 m!$, so the sum over such sets is the sum over ordered tuples of distinct components times $2^{-|\gamma(m)|}$, at most $2^{-|\gamma(m)|} (\sum_\tau 2^{-\text{bits}(\tau)})^m \leq 2^{-|\gamma(m)|}$; summing over m gives at most 1, and $(1 + |T|)^{-\tau} \leq 1$. An independent implementation of the code gives the same lengths to four decimals (experiments §1.3). (iii) Membership in a finite DT° theory is decided by matching against each template, in time $O(\sum_{\tau \in T} (|\tau| + |s|))$ (AS Thm 2.5); so the time constant is polynomial in $\ell(T)$ and its logarithm is $O(\ln \ell(T))$. $\square$

A.2 Instantiation grammar

Proof of Lemma 1.9. (a) Let G_j be the sum of h over generation j . Given the first j generations, $\mathbb{E}[G_{j+1}] \leq \rho G_j$ by the uniform condition and linearity, so $\mathbb{E}G_j \leq \rho^j h(i)$. Generation j has at most $G_j/h_{\min}$ nodes; sum over j .

(b) Let N_D count the nodes of depth at most D . We show $\mathbb{E}_j[e^{\zeta N_D}] \leq e^{\zeta \Lambda h(j)}$ for all types j by induction on D . For $D = 0$, $N_0 = 1 \leq \Lambda h_{\min} = 2/(1-\rho)$. For the step, let the root's children have types $c_1, \dots, c_r$ ($r \leq b$) and $H := \sum_l h(c_l) \in [0, bh_{\max}]$. The subtrees are independent given the types, so $\mathbb{E}_j[e^{\zeta N_{D+1}}] \leq e^{\zeta \mathbb{E}_j[\Lambda H]}$. Put $x := \zeta \Lambda H \in [0, 1]$ (as $\zeta \leq (\Lambda b h_{\max})^{-2}$ and $\Lambda b h_{\max} \geq 2$). For $x \in [0, 1]$, $e^x \leq 1 + x + x^2$, since $e^x - 1 - x \leq x^2(e-2)$. So $\mathbb{E}_j[e^{\zeta \Lambda H}] \leq 1 + \zeta \Lambda \rho h(j) + \zeta^2 \Lambda^2 b^2 h_{\max}^2 \leq \exp(\zeta \Lambda \rho h(j) + \zeta^2 \Lambda^2 b^2 h_{\max}^2)$, and $\mathbb{E}_j[e^{\zeta N_{D+1}}] \leq e^{\zeta \Lambda h(j)}$ iff $1 + \zeta \Lambda^2 b^2 h_{\max}^2 \leq \Lambda(1-\rho)h(j)$. The right side is at least 2, the left at most 2. Monotone convergence gives the bound for N ; the tail is Markov's inequality.

(c) By AS Thm 2.5(a), $\theta \mapsto \tau\theta$ is injective, so $\sum_s \mathcal{Q}_\tau(s) = \sum_\theta \mathcal{Q}(\theta) = \prod_M 1 = 1$, each body law being a probability on finite bodies by (a). With full support $\mathcal{Q}(\theta) > 0$ for every θ .

(d) The proofs of (a) and (b) use only independence given the type, the bound b and the uniform condition. $\square$

The per-type condition (Remark 1.10). The pre-referee definition asked that every type have mean offspring below 1, and dominated the process by one Galton–Watson process with the largest mean (referee M5). (1) No formula type of L_A or L_∞ satisfies this: every formula production has a child. (2) With infinitely many types it does not give finiteness: with a nullary predicate $\top$, let a formula node with k variables choose $\forall$ (child with $k+1$ variables) with probability $1 - 1/(k+2)^2$ and $\top$ otherwise; the chain never stops with probability $\prod_{m \geq 2} (1 - 1/m^2) = \frac{1}{2}$ (referee, `r1_subcritical.out`: product 0.500000, simulated 0.4989). (3) Means do not give domination: $\{0:0.55, 2:0.45\}$ and $\{0:0.1, 1:0.9\}$ both have mean 0.9, and a law dominating both has mean ≥ 1.35 . *The uniform condition excludes the chain of (2)* [proved]: it requires $h(k+1)(1 - 1/(k+2)^2) \leq \rho h(k)$, so $h(k) \leq h(0)\rho^k \prod_{j < k} \frac{(j+2)^2}{(j+1)(j+3)} = h(0)\rho^k \frac{2(k+1)}{k+2} \rightarrow 0$, against $h \geq h_{\min} > 0$ (`c10_subcritical.out`, Part C: e.g. $6.1 \cdot 10^{-92}$ at $k = 2000$ for $\rho = 0.9$).

Example A.1 (Grammars that satisfy Definition 1.8). `[computed]`^{model §1.4 (c10); experiments §1.2} $\mathcal{Q}_A$ for L_A : term node 0.4, S .2, $+$.1, $\cdot$.1, a variable .2 (uniform over the k variables; renormalised away when $k = 0$); formula node = .3, $<$.2, $\neg$, $\wedge$, $\rightarrow$, $\forall$, $\exists$.1 each. Term nodes have mean 0.75 ($k = 0$) or 0.6 term children; formula nodes 0.7 formula and 1.0 term children. With $h(o) = 1$, $h(\iota) = 0.05$ the weighted ratio is at most 0.75 at every type, so $\rho = 0.75$. From $2 \cdot 10^5$ bodies: mean size 14.62 (s.e. 0.04) from a formula root, against the bound 80; 4.016 (s.e. 0.015) from a term root at $k = 0$, against the bound 4, which is the exact mean $1/(1 - 0.75)$. Both tails decay exponentially. Track experiments' grammar has mean 0.72 same-sort children at formula nodes and at most 0.75 at term nodes, and satisfies the condition with $h(o) = 1$ and $h(\iota)$ small.

A.3 Calculi

One law $\mathcal{Q}$ instantiates metavariables and, unless said otherwise, supplies the $\forall E$ terms (universal §1.2).

$C_{\min}$. Chains. A node cites (probability $1 - c$: an axiom by weight, metavariables from $\mathcal{Q}$) or applies $\forall E$ (probability c) to a recursively generated premise with $t \sim \mathcal{Q}$, failing if the premise has no leading $\forall$. For $\{\sigma_\varphi\}$, $Z = (1-c) \sum_{k \leq m'} c^k$ with m' the number of leading $\forall$ of φ (??). $C_{\text{open}}(c, g)$. Adds Gen (probability g), which abstracts the parameter of the premise and fails if there is none;

$p_{\text{cite}} = 1 - c - g$; terms from $\mathcal{Q}_{\text{open}} = \rho[p] + (1 - \rho)\mathcal{Q}_{\text{num}}$ (ρ here is the parameter probability). Per closed datum, normalised: $H_{\forall}/H_{\text{sch}} = c(1 - \rho)/(1 + c)$, $H_{\text{open}}/H_{\text{sch}} = (1 - \rho)/(1 + g\rho)$ (??). $\mathcal{C}_{\cup 3}$. Each node picks cite, $\forall E$, or a rule whose premises and conclusion are quantifier-free (equality rules; propositional rules and MP on quantifier-free formulas), with fixed probabilities; trees. $\mathcal{C}_{\wedge}$. Cite (p_{cite}), $\forall E$ (c), $\wedge I$ (a), $\wedge E$ (e ; side left or right with probability $\frac{1}{2}$), the $\wedge$ -rules on all formulas; trees. Used for ??.

Ch_J (*experiments §1.6*). A chain $s_0, \dots, s_k$, $k \leq J$. s_0 cites component i with probability w_i (bodies from $\mathcal{Q}$, guards respected). At step $j < J$, if some rule applies, the chain stops with probability c_{stop} , otherwise it applies an applicable rule r with probability $\propto c_r$: *elim* ($s_j = \forall x \psi \mapsto \psi[t/x]$, t from $\mathcal{Q}_e$, closed terms, with p iff the run is open); *gen* ($s_j \mapsto \forall x s_j[x/p]$ if s_j contains p); *mp* (a uniformly chosen component $A \rightarrow B$, $A, B \in \text{DT}^\circ$ with the metavariables of A in B , whose A matches s_j ; $s_{j+1} = B\theta$, θ read off s_j and drawn from $\mathcal{Q}$ on the rest). If no rule applies, or $j = J$, it stops. Defaults $c_{\text{stop}} = \frac{1}{2}$, all $c_r = 1$, $J = 1$ ($J = 2$ in parts of E1 and in E6); one parameter p . The chain never fails, so its law is a probability (experiments Prop X3), and $P(s \mid T, w) = \sum_i w_i a_i(s)$. It is the special case of L1 with cited major premises and depth $\leq J$, so induction followed by MP and $\forall E$ is outside it (??).

ND and L1^{sch} (*pa Def 0.3*). Sequents $\Gamma \vdash A$ with rules hyp, ax (an axiom, or a schema instance at a motive), tc (from $\Gamma_j \vdash A_j$ infer $\bigcup_j \Gamma_j \vdash B$ if $\bigwedge_j A_j \rightarrow B$ is a tautology), $\rightarrow I$, $\forall E$, $\forall I$, $\exists I$, $\exists E$, refl, subst, with the usual side conditions; a standard sound and complete calculus, checked by `nd.py`. The derivation is written schematically in a predicate metavariable P and the motive is coded once by the grammar. Per line it costs $\log_2 10$ bits for the rule, $\log_2 |T|$ per axiom citation, $\log_2(\text{line index})$ per premise reference and $\beta = \log_2 23 \approx 4.52$ bits per written symbol. The decodable variant also codes the number of lines, the premise count of each tc line and the variables named by subst and $\exists I$ (4.8–7.3% more on pa §1’s derivations); it is a prefix code, so L1^{sch} is the two-part form of a sub-probability. (β should strictly be $\log_2$ of the alphabet size, about $\log_2 25$: about 3% on symbol costs, no conclusion changes.)

The grammar of L1^{eq} (*model Ex 3.6*). A node cites a template (α_{ax} ; metavariables from $\mathcal{Q}$) or applies refl ($t \sim \mathcal{Q}$), sym, trans (valid iff the middle terms agree), congruence for S , or congruence for $+$ (side term $u \sim \mathcal{Q}$, left or right). A tree is valid iff its steps are and all its terms are closed of size $\leq N$. μ_T is the exact least fixpoint; Monte Carlo agrees ($Z = 0.54909$ against 0.54965 ± 0.00091). The rules are complete for closed equational consequence (known; Birkhoff, 1935, not checked).

Proof sketch of Lemma 1.3(b) Let $s_{m,t} := \forall x B_m \rightarrow B_m[t/x]$, $B_m := (x + \dots + x = x)$ with m summands. Suppose $\tau \in \text{DT}^\circ$ with $\text{inst}(\tau) \subseteq \text{inst}(A4)$ covers $s_{m,t}$ and $s_{m',t'}$, $m \neq m'$. The antecedents differ under $\forall x$ at positions whose subterms contain the bound x , so τ has an occurrence there with x among its arguments, say $f(x)$. The matching position of the consequent carries $B_m[t/x]$; in τ it is either $f(c)$ with a fixed metavariable-free c , forcing $t = t' = c$ there, or a metavariable independent of f , which gives instances outside A4. So each template covers $s_{m,t}$ for one m or one t , and a finite set misses some $s_{m,t}$. *Not written out*: turning “the matching position in the consequent” into a statement about every DT° shape (model §10, open problem 9).

A.4 Likelihoods

Table 2 lists the likelihoods of Definitions 1.11, 1.13 and 1.15 and the scores of Definition 1.18.

Proof of Lemma 1.12. $\prod_i P_{T,w}(X_i) = \sum_\ell \prod_\tau w_\tau^{n_\tau(\ell)} \prod_i \mathcal{Q}_{\ell_i}(X_i)$, a finite sum over the labellings

name	definition	normaliser	size pr.	model / universal / pa / experiments
L0	$\sum_{\tau} w_{\tau} \mathcal{Q}_{\tau}(s)$, or P_T^{Dir}	probability	yes	L0 / L0-strict / L0 / L0
L0 ^{cl}	L0; $\forall \bar{x} \chi$ cited via instances	probability	yes [†]	— / L0-closure / — / —
μ_T	$\Pr[\text{valid tree concluding } s]$	mass Z_T	yes [‡]	μ_T / L1(c) / — / —
L1	μ_T / Z_T	$Z_T \geq 1 - \alpha_r$	yes	L1 / L1-norm / L1 (not computed) / L1 (Ch _J)
L2	citation forced at depth d	$Z_T^{(d)}$	yes	L2 / — / “bounded” / —
L1 ^σ	$L1 \cdot e^{-\kappa \pi }$, renormalised	Z_T^{σ}	yes	L1 ^σ / — / — / —
L1 ^{max}	best single tree	sub-prob.	yes [‡]	two-part / L1-max / L1-max / —
L1 ^{sch}	two-part ND code	sub-prob.	yes [‡]	— / — / L1-sch / —
L1 ^{sel}	$P_T(s)/P_T(S)$ on S	$P_T(S)$	yes [†]	— / L1-sel(S) / — / —
L1 ^{sel} _{cit}	$\sum_i w_i a_i(s)/c_i$	c_i	yes [†]	— / — / — / L1sel
P_T^{η}	$(1 - \eta)P_T + \eta N$	as P_T	yes [†]	(channel P_T^K) / P_T^{η} / — / —
L_{ε}	$(1 - \varepsilon)P_T + \varepsilon \mu_0[T \not\vdash_k \neg s]$	mass dep. on T	partly [§]	— / — / L_{ε} / —
L1 ^{eq}	equational grammar	Z_T	yes [†]	L1-eq / — / — / —
P_T^g	$2^{-\kappa \ell_T(s)} / Z_T$	$Z_T \leq 1$	restricted	P_T^g / — / — / —
scores	Definition 1.18	none	no	$S_{\text{prove}}, S_{\text{nc}}, S_g$ / $S_{\text{nc}, \beta}$ / L_{ε} / —

Table 2: Likelihoods and scores. Normalisers are data-independent. “yes”: Proposition 1.20(a); “restricted”: Remark 1.21; “no”: Proposition 1.20(b). [†]A fixed probability on $\mathcal{S}$, so Lemma 1.19 applies as in the proof of Proposition 1.20(a). [‡]A fixed sub-probability, so ?? applies. [§]Only $(1 - \varepsilon)P_T$ separates theories that refute no datum; at $\varepsilon = 1$ nothing is learned (??). The marks say that mass outside the data’s support is lost, not how strongly two theories are separated.

with $X_i \in \text{inst}(\ell_i)$; integrate term by term with the Dirichlet moments (known)

$$\mathbb{E} \prod_{\tau} w_{\tau}^{n_{\tau}} = \frac{\Gamma(A)}{\Gamma(A+n)} \prod_{\tau} \frac{\Gamma(\alpha_{\tau} + n_{\tau})}{\Gamma(\alpha_{\tau})}.$$

The result is symmetric in the data. P_T^{Dir} is a mixture of the i.i.d. laws $P_{T,w}^{\otimes n}$, each a probability on $\mathcal{S}^n$ consistent in n (Lemma 1.9(c)), so its normaliser is 1 for every D . With disjoint instance sets each datum has one label, and $P_T^{\text{Dir}}(D, s)/P_T^{\text{Dir}}(D) = \frac{\Gamma(A+n)}{\Gamma(A+n+1)} \frac{\Gamma(\alpha_{\tau} + n_{\tau} + 1)}{\Gamma(\alpha_{\tau} + n_{\tau})} \mathcal{Q}_{\tau}(s) = \frac{\alpha_{\tau} + n_{\tau}}{A+n} \mathcal{Q}_{\tau}(s)$ for $s \in \text{inst}(\tau)$. $\square$

Computed. `c8_formulas.out` (1): closed form $9.908294678 \cdot 10^{-6}$ against quadrature over $w_1 \sim \text{Beta}(\frac{1}{2}, \frac{1}{2})$ $9.908294680 \cdot 10^{-6}$ (asserted to 10^{-12}). Experiments Prop X2: the exact dynamic programme agrees with brute force on 500 random instances and with the referee’s independent count-vector sum on 400 cases to $1.4 \cdot 10^{-14}$; above 20000 states it returns rigorous bounds (??).

Proof of Lemma 1.14. (a) A derivation node has 0, 1 or 2 premises with probabilities $\alpha_{\text{ax}} + \alpha_{\text{lg}} > 0$, $\alpha_{\text{Gen}} + \alpha_{\text{vE}}$, α_{MP} , independently: a Galton–Watson tree with mean m . The extinction theorem (known) gives finiteness iff $m \leq 1$, given positive mass at 0 (without it the tree is infinite even at $m = 1$; referee m11), and extinction probability q for $m > 1$. $\mathbb{E}N = 1 + m\mathbb{E}N$, with $\mathbb{E}N < \infty$ for $m < 1$ since generation k has mean m^k . Bodies and terms are finite a.s. by Lemma 1.9(a). (b) A one-node citation tree is valid, so $Z_T \geq \alpha_{\text{ax}} + \alpha_{\text{lg}}$. (c) $\subseteq$: a valid tree is a K-derivation from $\bigcup \text{inst}(T)$, each $\forall E$ node replaced by an A4 instance and MP. $\supseteq$: a K-derivation from $\bigcup \text{inst}(T)$ and logical axioms is a grammar tree of positive probability: citations have $w_{\tau} \mathcal{Q}(\theta) > 0$, logical-axiom instances (A4’s term included) have positive $\mathcal{Q}$ -probability with parameters where used, and each Gen has positive parameter probability. Without parameters all formulas in a tree are closed and Gen adds only vacuous quantifiers (referee m12). $\square$

Implicit in the notes. For $T = \emptyset$ no theory axiom can be cited; reading such a node as invalid, (b) becomes $Z_\emptyset \geq \alpha_{\text{lg}} > 0$, which is what Proposition 1.22(b) uses.

Computed (c8_formulas.out (2), derivation nodes): $(\alpha_{\text{MP}}, \alpha_{\text{Gen}} + \alpha_{\text{VE}}) = (0.15, 0.2)$, $m = 0.5$: mean 2.010 against 2; $(0.3, 0.3)$, $m = 0.9$: 9.965 against 10; $m = 1$: heavy tail; $(0.4, 0.3)$, $m = 1.1$: $P(\text{finite}, < 20000 \text{ nodes}) = 0.7489$ against $q = 0.75$.

Proof of Lemma 1.16 and Remark 1.17. A one-node citation tree π has $\Pr(\pi) e^{-\kappa|\pi|} > 0$, and $e^{-\kappa|\pi|} \leq 1$ gives $\mu_T^\sigma \leq \mu_T$; $\mu_T^\sigma(s) > 0$ iff some valid tree concludes s with positive probability, iff $\mu_T(s) > 0$. For Remark 1.17: with one component $P_T = a_1$ and $P_T(S) = c_1$; with several, the stream filter divides by $\sum_i w_i c_i$ and the per-citation form each term by its own c_i . $\square$

A.5 The size principle

Proof of Lemma 1.19. On $\text{supp } P^*$, $P = (1 - \varepsilon)\tilde{P}$; substitute into

$$\text{KL}(P^* \| P) = \sum_{s \in \text{supp } P^*} P^*(s) \ln(P^*(s)/P(s)).$$

For (b), $\mathbb{E}_{P^*}[\tilde{P}(X)/P^*(X)] = \sum_{s \in \text{supp } P^*} \tilde{P}(s) = 1$; extend to n data by independence. $\square$

Computed (c1_size_principle.out): root law $(0, S, +, \cdot) = (.5, .3, .1, .1)$; P^* the law of $\{z + 0 = z\}$, P that of $\{z_1 + 0 = z_2\}$ on the 30901 terms of size ≤ 11 . (a) holds to 10^{-9} (asserted); the simulated per-datum log-ratio is 3.81, 4.19, 3.94 nats at $n = 10^2, 10^3, 10^4$, against $\text{KL} = H(\mathcal{Q}) = 3.894$ nats for the untruncated grammar.

Proof of Proposition 1.20. (a) Each P_T is a fixed probability on $\mathcal{S}$ (Lemmas 1.9, 1.14 and 1.16; under L2 all trees are finite and one-node citations valid); apply Lemma 1.19. Under L0 with full support and positive weights, $\text{supp } P_{T,w} = \bigcup \text{inst}(T)$. (b) A model of $T' \cup D_+ \cup \neg D_-$ is a model of $\text{Th}(T') \supseteq \bigcup \text{inst}(T)$, so $S_{\text{nc}}(T') \leq S_{\text{nc}}(T)$. If $S_{\text{prove}}(T) = 1$, T' proves what T proves and is consistent by hypothesis, so $S_{\text{prove}}(T') = 1$. For S_g : $S_{\text{nc}}(T') = 1$ by hypothesis, and instance inclusion makes every derivation from T one from T' , so $\ell_{T'} \leq \ell_T$ and $g(\ell_{T'}) \geq g(\ell_T)$. The consequence is the case $T = T^*$, $T' = T^* \cup \{\psi\}$, where the instances are nested; if T^* proves the labelled data, the consistency hypothesis reduces to that of $T^* \cup \{\psi\}$. (c) Under L0, $P_{T'}(s) = (1 - w_\psi)P_{T^*}(s) + w_\psi[s = \psi]$. Under L1, the one-node tree citing ψ gives $P_{T'}(\psi) \geq \alpha_{\text{ax}} w_\psi$ (as $Z_{T'} \leq 1$), while $\psi \notin \text{Th}(T^*) \supseteq \text{supp } P_{T^*}$ by the inclusion $\subseteq$ of Lemma 1.14(c), which needs no hypothesis. So $\varepsilon \geq \alpha_{\text{ax}} w_\psi$ in Lemma 1.19, and by the strong law $n^{-1} \ln \frac{P_{T'}(D)}{P_{T^*}(D)} \rightarrow -\text{KL}(P_{T^*} \| P_{T'}) \leq -\ln \frac{1}{1 - \alpha_{\text{ax}} w_\psi}$ a.s. The Dirichlet case is ??(a). $\square$

Proof of Remark 1.21. A derivation determines its conclusion, so distinct sentences have distinct shortest derivations with codes in one prefix-free set; with $\kappa \geq 1$, Kraft gives $Z_T \leq 1$. With $g_\infty > 0$ the infinitely many sentences without derivation make the sum infinite. The code writes axioms as formulas, so instance inclusion makes every derivation from T one from T' with the same code: $\ell_{T'} \leq \ell_T$, $Z_{T'} \geq Z_T$. The refuted sentence fails where the numerator grows too. The counterexample is the referee's (r6_stronger_theory.out): on a universe of 570 formulas of size ≤ 7 , $\ell_T(b) = 5$ and $\ell_{T'}(b) = 1$; $\kappa = 1$ gives $P_T(b) = 0.0321$, $P_{T'}(b) = 0.3456$ ($\kappa = \frac{1}{2}$: 0.0432, 0.1491), 10 sentences gaining and 36 losing; under L2 (depth 2) $P_T(b)$ rises from 0.0134, 0.0156, 0.0168 to 0.263, 0.240, 0.196 in three rule settings. $\square$

A.6 Computing the likelihood

Proof of Proposition 1.22. (a) Let $\nu(\pi)$ count the grammar nodes of a tree: derivation nodes, $\mathcal{Q}$ nodes of bodies and terms, and the nodes of the geometric chains that pick parameters. Weight them 1, $\eta h_{\mathcal{Q}}$ and η . A derivation node then has expected weighted offspring at most $m + \eta(v_T h_{\mathcal{Q}, \max} + 1)$, with v_T the largest number of metavariable slots of a template of T or of a logical schema; a $\mathcal{Q}$ node has ratio at most $\rho_{\mathcal{Q}}$; a chain node continues with a fixed probability below 1. As $m < 1$, a small η makes all ratios less than 1, so the whole tree satisfies Lemma 1.9(d) (as in the proof of ??) and $\mathbb{E}\nu \leq C_\nu$, computable from the parameters and T . Finitely many trees have $\nu \leq N$ (each node chooses from a finite list: rule, template, logical schema, production, one of at most the arity plus N variables, chain step); validity is decidable and each probability is a computable real. So $\mu_T^{\leq N}(s) \leq \mu_T(s) \leq \mu_T^{\leq N}(s) + C_\nu/N$ by Markov's inequality, likewise for Z_T , and $Z_T \geq 1 - \alpha_r > 0$. (b) The identity is Lemma 1.14(c). $\bigcup \text{inst}(T)$ is decidable by matching, so $\text{Th}(T)$ is recursively enumerable. $\text{Th}(\emptyset)$ is the set of valid formulas, undecidable for a language with a binary relation symbol (Church, Turing; known). (c) Deciding $P_T(s) = 0$ would decide $s \notin \text{Th}(T)$. For a finite class, $\pi_n(T)$ is a quotient of computable reals whose denominator is positive when defined. $\square$

Proof of Proposition 1.23. For a body β of type $\iota^2 \rightarrow o$ let β^{sw} swap its holes. Then $\tau'[F := \beta] = \tau[F := \beta^{\text{sw}}]$, so $\text{inst}(\tau) = \text{inst}(\tau')$, and hole symmetry gives $\mathcal{Q}_\tau = \mathcal{Q}_{\tau'}$. Under L0, $P_{T_2} = \frac{1}{2}\mathcal{Q}_\tau + \frac{1}{2}\mathcal{Q}_{\tau'} = P_{T_1}$, and the posterior odds are the prior odds (?). The codes of τ and τ' differ, so $T_1 \neq T_2$. Under the maximum each instance has exactly one matcher per template (AS Thm 2.5), so two citations of probability $\frac{1}{2}\mathcal{Q}_\tau(s)$ each, and $V_{T_2}(s) = \frac{1}{2}V_{T_1}(s)$. $\square$

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